EMS Gateway API, Data Model and Integration Specification

Stable Hardware-Verified Baseline - v1.0

Generated from the uploaded kinetics_grid_gateway_stable_baseline(2).zip codebase.
Audience: Frontend Team, Backend/API Team, Database Team, Dashboard Team, EMS Control Team

Item	Value
Source package	kinetics_grid_gateway_stable_baseline(2).zip
Gateway folder analyzed	imx93_gateway/
Main API port	8000 - EMS Web API
Log API port	7000 - HTTP Log API
TCP command port	6000 - TCP Command Server
Document purpose	Field-level handoff contract for frontend/backend/database development

1. Executive Overview

The EMS Gateway is an i.MX93 based integration backend that connects physical assets and simulators to web dashboards, Flutter/Desktop dashboards, historical log storage and future EMS control logic.

This document is based on the actual code structure of the stable baseline. It describes the REST APIs, log APIs, data fields, response envelopes, storage schemas, command model, health model and database integration recommendations exposed by the gateway.

The gateway is intentionally modular: asset communication is separated from web APIs, telemetry composition, command dispatching, storage/query handling, health monitoring and operator telemetry filtering.

2. Runtime Architecture and Network Ports

The stable baseline includes a small main.py entrypoint, an application orchestration layer, asset adapters, protocol descriptors, storage/query abstractions, health monitoring, dynamic asset runtime catalog and operator telemetry view.

The operating network model used during hardware testing is eth0 for Flutter/Desktop/PC side, eth1 for PCS/Inverter side and mlan0 for Wi-Fi Web Dashboard access.

The gateway exposes three major network services: TCP command server on port 6000, HTTP Log API on port 7000 and EMS Web API on port 8000.

Service	Default Port	Primary Consumer	Purpose
EMS Web API	8000	Web dashboard, backend services	Asset catalog, latest telemetry, operator telemetry, health, diagnostics and commands
HTTP Log API	7000	Backend/database, dashboards, reporting tools	Historical telemetry/event/error logs, storage health and CSV download
TCP Command Server	6000	Flutter/Desktop dashboard	Legacy TCP command execution and telemetry/status commands
UDP Telemetry Streamer	5005 destination	Flutter/Desktop dashboard	Realtime legacy-compatible telemetry broadcast to PC side

3. Folder and Module Map

Folder / file	Role in gateway	Why frontend/backend teams should care
main.py	Small entrypoint that parses CLI and starts EMSGatewayApplication	Stable startup point; no API logic should be added here
core/app/gateway_application.py	Service orchestration and callback source for APIs	Runtime status/telemetry/command callbacks originate here
network/ems_web_api_server.py	REST API and SSE server on port 8000	Primary source for frontend/backend API contracts
network/log_http_server.py	HTTP Log API server on port 7000	Primary source for historical log/filter API contracts
core/assets/runtime_catalog.py	Dynamic asset catalog and RuntimeAssetRecord model	Defines fields exposed by /api/assets
core/operator_telemetry_view.py	Operator telemetry filter	Defines what dashboard operator view hides/preserves
core/health/health_monitor.py	Health and diagnostics response builder	Defines /api/health and /api/diagnostics response model
core/storage/query/*	LogFilter, LogQueryResult and CSVLogQueryBackend	Defines log filters and portable query abstraction
services/storage_logger.py	CSV storage backend and log schemas	Defines telemetry/event/error CSV fields

models/*.py	EMS-facing asset state models	Defines major data dictionaries for PCS/BMS/Chiller
drivers/bms_register_map.py	BMS register and control map	Defines BMS decoded signal source and control registers
drivers/pcs_profiles/*	PCS vendor profiles	Defines vendor-specific register maps/scaling
configs/*.json	Runtime profiles	Defines IPs, ports, asset profile config and deployment modes

4. EMS Web API Endpoint Inventory - Port 8000

These endpoints are exposed by network/ems_web_api_server.py. They are the primary endpoints for the frontend dashboard and backend integration services.

Method	Endpoint	Purpose	Query/body parameters	Response fields
GET	/	API discovery root with endpoint list and server metadata		status, server, message, timestamp, endpoints
GET	/api/gateway/health	Lightweight Web API server liveness check		status, server, message, timestamp, host, port
GET	/api/health	Full overall gateway health model		status, timestamp, gateway_id, gateway, assets, summary
GET	/api/health/gateway	Gateway/service-level health		status, gateway_id, timestamp, mode, summary, services, recommended_action
GET	/api/health/assets	All asset health map		status, timestamp, summary, assets
GET	/api/health/assets/{asset_id}	Single asset health		status, asset_id, asset_key, asset_type, enabled, running, online, protocol, profile, vendor, connection, last_successful_poll, last_error, consecutive_failures, reason, recommended_action, storage
GET	/api/diagnostics	Gateway and all asset diagnostics		status, timestamp, gateway, diagnostics
GET	/api/diagnostics/assets/{asset_id}	Single asset diagnostics		status, timestamp, asset_id, diagnostics, health
GET	/api/gateway/status	Full runtime status packet		gateway_id, mode, chiller, pcs, bms, network, web_api, log_http, runtime_config, configured_assets, asset_factory_plan
GET	/api/gateway/network	Network endpoint summary for frontend/backends		gateway_id, tcp_command, udp_telemetry, log_http, web_api, note
GET	/api/assets	Runtime asset catalog from configured assets + active +		status, gateway_id, timestamp, assets_count, assets,

		latest telemetry		summary
GET	/api/assets/{asset_id}	Single runtime asset record		status, asset, timestamp
GET	/api/telemetry/latest	Latest full engineering telemetry packet	view=operator optional	type, gateway_id, asset_id, timestamp, data, pcs, bms, assets
GET	/api/telemetry/operator	Operator-filtered telemetry packet		Same as telemetry/latest but raw/debug/storage fields removed and view=operator
GET	/api/assets/{asset_id}/ telemetry/latest	Latest engineering telemetry for one asset	view=operator optional	status, asset_id, asset_type, asset_key, timestamp, online, runtime_mode, telemetry
GET	/api/assets/{asset_id}/ telemetry/operator	Operator-filtered latest telemetry for one asset		status, asset_id, asset_type, asset_key, timestamp, online, runtime_mode, telemetry, view
GET	/api/assets/{asset_id}/ telemetry/keys	Flattened telemetry key list for dynamic frontend screens	view=operator optional	status, asset_id, asset_type, timestamp, keys_count, keys
GET	/api/assets/{asset_id}/ telemetry/timeseries	Timeseries from log files for charting	key or keys, date, limit	status, asset_id, asset_type, timestamp, keys, date, points
GET	/api/stream/telemetry	Server-Sent Events stream for live dashboard	view=operator optional	SSE event: telemetry
POST	/api/commands	Generic command execution endpoint	JSON body: command, asset_id or asset_type, request_id, value/parameters	status, timestamp, asset_id, command, request_id, result
POST	/api/assets/{asset_id}/ commands	Asset-scoped command execution endpoint	JSON body: command, request_id, value/parameters	status, timestamp, asset_id, command, request_id, result

5. HTTP Log API Endpoint Inventory - Port 7000

These endpoints are exposed by network/log_http_server.py. They are intended for historical logs, reporting, database ingestion and CSV export.

Method	Endpoint	Purpose	Query parameters	Response fields
GET	/	Log API discovery root		status, server, message, endpoints
GET	/api/health	Log HTTP server liveness		status, server, timestamp
GET	/api/logs/assets	List assets with log folders/files		status, base_path, assets
GET	/api/storage/status	Storage/log file status	asset_id optional	status, asset_id, base_path, telemetry_dir_exists, files, sizes, disk fields
GET	/api/storage/health	Operator/diagnostic storage health	asset_id optional	status, message, asset_id, base_path, disk_total_bytes, disk_used_bytes, disk_free_bytes, disk_used_percent, log_total_bytes

GET	/api/logs/files	List telemetry/event/error files for asset	asset_id optional	status, asset_id, telemetry_files, event_file, error_file
GET	/api/logs/telemetry	Query historical telemetry logs	asset_id, date required, start_time, end_time, fields, exact filters, search, limit, offset, order	status, log_type, asset_id, date, rows, total_rows, filtered_rows, rows_count, filters
GET	/api/logs/events	Query event logs	asset_id, event_type, status, source, vendor, command, search, fields, date/time, limit, offset, order	status, log_type, asset_id, rows, total_rows, filtered_rows, rows_count, filters
GET	/api/logs/errors	Query error logs	asset_id, error_type, error_source or source, search, fields, date/time, limit, offset, order	status, log_type, asset_id, rows, total_rows, filtered_rows, rows_count, filters
GET	/api/logs/metadata	Gateway metadata text from storage		status, metadata_file, content
GET	/api/logs/download/telemetry	Download telemetry CSV	asset_id, date required	text/csv attachment

6. Runtime Asset Catalog Field Dictionary

The /api/assets endpoint is based on RuntimeAssetCatalog and RuntimeAssetRecord. Frontend code should use this endpoint for asset discovery instead of hardcoding PCS/BMS/Chiller cards.

Field	Type	Default	Meaning
asset_id	str		Unique asset identifier, for example pcs_1, bms_1, chiller_1.
asset_key	str		Normalized key used internally for routing, for example pcs/bms/chiller.
asset_type	str		Logical asset type such as pcs, bms, chiller or future asset type.
enabled	bool	False	Whether asset is enabled by config/CLI.
running	bool	False	Whether active runtime service is currently running.
online	bool	False	Whether latest telemetry/status indicates online communication.
protocol	Optional[str]	None	Communication protocol such as modbus_tcp, modbus_rtu, can or mock.
profile	Optional[str]	None	Vendor/profile name used for register map and scaling.
vendor	Optional[str]	None	Vendor/profile family.
configured	bool	False	Runtime asset catalog field exposed in /api/assets responses.
telemetry_available	bool	False	Whether latest telemetry exists for this asset.
runtime_mode	str	'configured_only'	active_service, configured_only, configured_future or disabled.
compatibility	JsonDict		Runtime asset catalog field exposed in /api/assets

			responses.
connection	JsonDict		Connection metadata like host/port/unit_id or serial settings.
metadata	JsonDict		Free-form extra asset metadata.

7. /api/assets - Runtime Asset Catalog

Item	Details
Endpoint	GET /api/assets
Purpose	Returns every configured/active asset after merging asset config, runtime status and latest telemetry. Use it to create dynamic asset cards.
Important response fields	status, gateway_id, timestamp, assets_count, assets, summary
Recommended consumer	Web dashboard / backend integration service

8. /api/assets/{asset_id} - Asset Details

Item	Details
Endpoint	GET /api/assets/pcs_1
Purpose	Returns one RuntimeAssetRecord. Use for asset detail screens and dynamic configuration displays.
Important response fields	status, asset, timestamp
Recommended consumer	Web dashboard / backend integration service

9. /api/telemetry/latest - Full Engineering Telemetry

Item	Details
Endpoint	GET /api/telemetry/latest
Purpose	Returns the combined latest engineering telemetry packet. It may include raw/debug/storage fields and is better for backend/engineering tools than operator dashboard grids.
Important response fields	type, gateway_id, asset_id, timestamp, data, pcs, bms, assets
Recommended consumer	Web dashboard / backend integration service

10. /api/telemetry/operator - Operator Telemetry

Item	Details
Endpoint	GET /api/telemetry/operator
Purpose	Returns telemetry filtered for operational dashboards. It removes raw register arrays, storage_logger, *_raw fields and internal/debug fields.
Important response fields	view, type/status, gateway_id, asset_id, timestamp, data, pcs, bms, assets
Recommended consumer	Web dashboard / backend integration service

11. /api/assets/{asset_id}/telemetry/latest

Item	Details
Endpoint	GET /api/assets/bms_1/telemetry/latest

Purpose	Returns latest telemetry for a selected asset. This is useful for asset detail cards and backend ingestion of single asset data.
Important response fields	status, asset_id, asset_type, asset_key, timestamp, online, runtime_mode, telemetry
Recommended consumer	Web dashboard / backend integration service

12. /api/assets/{asset_id}/telemetry/keys

Item	Details
Endpoint	GET /api/assets/pcs_1/telemetry/keys?view=operator
Purpose	Returns flattened telemetry keys. Use this for dynamic table/chart field pickers.
Important response fields	status, asset_id, asset_type, timestamp, keys_count, keys
Recommended consumer	Web dashboard / backend integration service

13. /api/assets/{asset_id}/telemetry/timeseries

Item	Details
Endpoint	GET /api/assets/pcs_1/telemetry/timeseries?keys=active_power_kw,dc_voltage_v&date=YYYY-MM-DD&limit=100
Purpose	Builds chart-ready points from CSV logs through LogQueryService. This endpoint is suitable for trend charts.
Important response fields	status, asset_id, asset_type, timestamp, keys, date, points
Recommended consumer	Web dashboard / backend integration service

14. /api/stream/telemetry - SSE

Item	Details
Endpoint	GET /api/stream/telemetry?view=operator
Purpose	Streams latest telemetry as Server-Sent Events. Use for live browser dashboards where polling is undesirable.
Important response fields	event: telemetry, data: JSON telemetry packet
Recommended consumer	Web dashboard / backend integration service

15. Operator Telemetry Filtering Contract

Operator telemetry is generated by core/operator_telemetry_view.py. This filter allows the gateway to keep full engineering telemetry while exposing a clean dashboard view.

Hidden exact key
control_mode_raw
debug
diagnostics
fault_active_bits
fault_binary
internal
on_off_raw
on_off_status_raw
raw_log_response
raw_registers
raw_telemetry_registers
register_map
set_temperature_raw

settings_registers_200_to_208
storage_logger

Hidden key substring
raw_telemetry
raw_register
registers_
_registers
register_block
storage_logger
logger_
_logger
debug
internal
binary
active_bits

Allowed top-level key
asset_id
asset_type
assets
bms
chiller
data
gateway_id
last_poll_time
message
online
pcs
status
telemetry
timestamp
type

16. Health and Diagnostics API Data Model

Health APIs are built by core/health/health_monitor.py. Health is distinct from telemetry: telemetry tells what values are; health tells whether the gateway can trust those values and what action should be taken.

Field	Meaning
status	Status string such as ok, error, healthy, degraded, offline, disabled.
gateway_id	Identifier of the gateway instance.
timestamp	ISO timestamp for the sample or response.
mode	Health/diagnostics response field generated by HealthMonitor.
summary	Health/diagnostics response field generated by HealthMonitor.
services	Health/diagnostics response field generated by HealthMonitor.
recommended_action	Action for operator/backend team to troubleshoot an unhealthy state.
asset_id	Unique asset identifier, for example pcs_1, bms_1, chiller_1.
asset_key	Normalized key used internally for routing, for example pcs/bms/chiller.
asset_type	Logical asset type such as pcs, bms, chiller or future asset type.
enabled	Whether asset is enabled by config/CLI.
running	Whether active runtime service is currently running.
online	Whether latest telemetry/status indicates online

	communication.
protocol	Communication protocol such as modbus_tcp, modbus_rtu, can or mock.
profile	Vendor/profile name used for register map and scaling.
vendor	Vendor/profile family.
connection	Connection metadata like host/port/unit_id or serial settings.
last_successful_poll	Timestamp of last successful communication if available.
last_error	Latest communication, storage or service error.
consecutive_failures	Failure count used by diagnostics/health layers.
reason	Human-readable reason for health state.
storage	Nested storage health/status for the asset.

Status value	Meaning	Frontend color recommendation
healthy	Asset/service is running and latest state is acceptable	Green
degraded	Service may be running but telemetry/storage/error status indicates warning/error	Orange/Yellow
offline	Enabled asset/service is not running or communication failed	Red
disabled	Asset/service disabled by config/CLI	Grey
unknown	Not enough information available yet	Grey

17. Command API and Command Dictionary

Commands can be sent through POST /api/commands or POST /api/assets/{asset_id}/commands. The command dispatcher routes packets based on command prefix, asset_id or asset_type.

Asset route	Command(s)	Meaning	Notes
Gateway	STATUS, GATEWAY_STATUS	Read gateway status packet	No write to physical asset
Gateway	READ_ALL_ASSETS, READ_GATEWAY_TELEMETRY	Read combined telemetry packet	No write to physical asset
PCS	PCS_READ	Read PCS status/telemetry	Routed to PCS asset
PCS	PCS_STATUS	Read PCS status/telemetry	Routed to PCS asset
PCS	READ_PCS	Read PCS status/telemetry	Routed to PCS asset
PCS	PCS_POWER_ON	PCS control command.	Logged as PCS event
PCS	PCS_POWER_OFF	PCS control command.	Logged as PCS event
PCS	PCS_SET_ACTIVE_POWER	PCS write/control command. SET_ACTIVE_POWER expects active power value; SET_REACTIVE_POWER expects reactive power value.	Logged as PCS event
PCS	PCS_SET_REACTIVE_POWER	PCS write/control command. SET_ACTIVE_POWER expects active power value; SET_REACTIVE_POWER expects reactive power value.	Logged as PCS event
PCS	PCS_RESET_FAULT	PCS write/control command. SET_ACTIVE_POWER expects active power value; SET_REACTIVE_POWER expects reactive power value.	Logged as PCS event
PCS	PCS_HEARTBEAT	PCS control command.	Logged as PCS event
PCS	PCS_STANDBY	PCS control command.	Logged as PCS event
BMS	READ_BMS	BMS read/control command	Routed to BMS service/driver
BMS	READ_BMS_ALARMS	BMS read/control command	Routed to BMS service/driver
BMS	READ_BMS_ALL	BMS read/control command	Routed to BMS service/driver
BMS	RESET_BCU	BMS read/control command	Routed to BMS service/driver

BMS	RESET_BMS	BMS read/control command	Routed to BMS service/driver
BMS	START_BMS_INSULATION_TEST	BMS read/control command	Routed to BMS service/driver
BMS	START_BMS_PRECHARGE	BMS read/control command	Routed to BMS service/driver
BMS	START_INSULATION_TEST	BMS read/control command	Routed to BMS service/driver
BMS	STOP_BMS_PRECHARGE	BMS read/control command	Routed to BMS service/driver
BMS	BMS_FAN_AUTO	BMS fan control wrapper command	Available through BMS service convenience methods
BMS	BMS_FAN_ON	BMS fan control wrapper command	Available through BMS service convenience methods
BMS	BMS_FAN_OFF	BMS fan control wrapper command	Available through BMS service convenience methods
Chiller	READ_ALL	Chiller read/write command	Default legacy route for dashboard commands
Chiller	READ_TEMP	Chiller read/write command	Default legacy route for dashboard commands
Chiller	READ_ONOFF	Chiller read/write command	Default legacy route for dashboard commands
Chiller	READ_MODE	Chiller read/write command	Default legacy route for dashboard commands
Chiller	READ_SETTINGS	Chiller read/write command	Default legacy route for dashboard commands
Chiller	SET_TEMP	Chiller read/write command	Default legacy route for dashboard commands
Chiller	SET_MODE	Chiller read/write command	Default legacy route for dashboard commands
Chiller	CHILLER_ON	Chiller read/write command	Default legacy route for dashboard commands
Chiller	CHILLER_OFF	Chiller read/write command	Default legacy route for dashboard commands

Request field	Required	Meaning
command	Yes	Command name normalized to uppercase before dispatch
asset_id	Required for /api/commands, injected by /api/assets/{asset_id}/commands	Target asset id such as pcs_1 or bms_1
asset_type	Alternative to asset_id for generic command route	Target type such as pcs/bms/chiller
request_id	Optional	Correlation identifier; generated if absent
value	Command-dependent	Setpoint value for commands such as SET_TEMP or PCS_SET_ACTIVE_POWER
client/source	Auto-added by API server	Source IP and web_api source string

18. Chiller Telemetry Field Dictionary

Field	Type	Default	Meaning
water_pump	Optional[str]	None	Decoded chiller engineering telemetry field.
compressor1	Optional[str]	None	Decoded chiller engineering telemetry field.
compressor2	Optional[str]	None	Decoded chiller engineering telemetry field.
electric_heater	Optional[str]	None	Decoded chiller engineering telemetry field.
condensate_fan	Optional[str]	None	Decoded chiller engineering telemetry field.
makeup_pump	Optional[str]	None	Decoded chiller engineering telemetry field.

outlet_water_temp	Optional[float]	None	Decoded chiller engineering telemetry field.
return_water_temp	Optional[float]	None	Decoded chiller engineering telemetry field.
outlet_water_pressure	Optional[float]	None	Decoded chiller engineering telemetry field.
return_water_pressure	Optional[float]	None	Decoded chiller engineering telemetry field.
ambient_temp	Optional[float]	None	Decoded chiller engineering telemetry field.
fault_code	Optional[int]	None	Decoded chiller engineering telemetry field.
control_mode	Optional[int]	None	Decoded chiller engineering telemetry field.
set_temperature	Optional[float]	None	Decoded chiller engineering telemetry field.
communication_status	<class 'str'>	'unknown'	Decoded chiller engineering telemetry field.
last_update_time	Optional[str]	None	Decoded chiller engineering telemetry field.

19. PCS Telemetry Field Dictionary

Field	Type	Default	Meaning
asset_id	<class 'str'>	'pcs_1'	Unique asset identifier, for example pcs_1, bms_1, chiller_1.
vendor	<class 'str'>	'njoy'	Vendor/profile family.
comm_status	<class 'str'>	'offline'	Vendor-independent PCS/Inverter telemetry or status field.
last_update_ts	<class 'str'>	''	Vendor-independent PCS/Inverter telemetry or status field.
error	<class 'str'>	''	Vendor-independent PCS/Inverter telemetry or status field.
ab_voltage_v	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
bc_voltage_v	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
ca_voltage_v	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
phase_a_voltage_v	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
phase_b_voltage_v	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
phase_c_voltage_v	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
phase_a_current_a	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
phase_b_current_a	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
phase_c_current_a	Optional[float]	None	Vendor-independent

			PCS/Inverter telemetry or status field.
frequency_hz	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
active_power_kw	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
reactive_power_kvar	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
apparent_power_kva	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
power_factor	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
bus_voltage_v	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
battery_voltage_v	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
battery_current_a	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
dc_power_kw	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
dc_total_current_a	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
operating_status_raw	Optional[int]	None	Vendor-independent PCS/Inverter telemetry or status field.
operating_status	<class 'str'>	'unknown'	Vendor-independent PCS/Inverter telemetry or status field.
grid_offgrid_status_raw	Optional[int]	None	Vendor-independent PCS/Inverter telemetry or status field.
grid_offgrid_status	<class 'str'>	'unknown'	Vendor-independent PCS/Inverter telemetry or status field.
fault_status	<class 'bool'>	False	Vendor-independent PCS/Inverter telemetry or status field.
hardware_fault_word_1_raw	Optional[int]	None	Vendor-independent PCS/Inverter telemetry or status field.
hardware_fault_word_2_raw	Optional[int]	None	Vendor-independent PCS/Inverter telemetry or status field.
grid_fault_word_raw	Optional[int]	None	Vendor-independent PCS/Inverter telemetry or status field.
bus_fault_word_raw	Optional[int]	None	Vendor-independent PCS/Inverter telemetry or status field.
ac_capacitor_fault_word_raw	Optional[int]	None	Vendor-independent PCS/Inverter telemetry or status field.
system_fault_word_raw	Optional[int]	None	Vendor-independent PCS/Inverter telemetry or status field.

switch_fault_word_raw	Optional[int]	None	Vendor-independent PCS/Inverter telemetry or status field.
other_fault_word_raw	Optional[int]	None	Vendor-independent PCS/Inverter telemetry or status field.
fault_words_raw	Dict[str, Any]	factory/default list or dict	Vendor-independent PCS/Inverter telemetry or status field.
fault_categories	Dict[str, Any]	factory/default list or dict	Vendor-independent PCS/Inverter telemetry or status field.
active_faults	List[str]	factory/default list or dict	Vendor-independent PCS/Inverter telemetry or status field.
fault_count	<class 'int'>	0	Vendor-independent PCS/Inverter telemetry or status field.
detailed_fault_status	<class 'bool'>	False	Vendor-independent PCS/Inverter telemetry or status field.
fault_words_read_error	<class 'str'>	''	Vendor-independent PCS/Inverter telemetry or status field.
igbt_temperature_c	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
ambient_temperature_c	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
inductance_temperature_c	Optional[float]	None	Vendor-independent PCS/Inverter telemetry or status field.
raw_telemetry	Dict[str, Any]	factory/default list or dict	Vendor-independent PCS/Inverter telemetry or status field.

20. BMS Telemetry Field Dictionary

Field	Type	Default	Meaning
gateway_id	str	'imx93_gateway_1'	Identifier of the gateway instance.
asset_id	str	'bms_1'	Unique asset identifier, for example pcs_1, bms_1, chiller_1.
timestamp	str	factory/default list or dict	ISO timestamp for the sample or response.
communication_status	str	'unknown'	EMS-facing BMS/BCU telemetry, health, alarm or status field.
last_error	Optional[str]	None	Latest communication, storage or service error.
last_success_ts	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
soc_percent	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
soh_percent	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
rack_inner_soc_percent	Optional[float]	None	EMS-facing BMS/BCU

			telemetry, health, alarm or status field.
rack_voltage_v	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
rack_current_a	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
power_kw	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
max_allowed_charge_current_a	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
max_allowed_discharge_current_a	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
max_cell_voltage_mv	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
min_cell_voltage_mv	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
avg_cell_voltage_mv	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
cell_voltage_diff_mv	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
max_cell_temp_c	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
min_cell_temp_c	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
avg_temp_c	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
max_temp_diff_c	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
insulation_resistance_kohm	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
positive_insulation_resistance_kohm	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
negative_insulation_resistance_kohm	Optional[float]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
precharge_stage	Optional[str]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
bcu_state	Optional[str]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
current_state	Optional[str]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
heartbeat	Optional[int]	None	EMS-facing BMS/BCU telemetry, health, alarm or status field.
contactor_active_flags	List[str]	factory/default list or dict	EMS-facing BMS/BCU telemetry, health, alarm or status field.

positive_contactor_closed	bool	False	EMS-facing BMS/BCU telemetry, health, alarm or status field.
precharge_contactor_closed	bool	False	EMS-facing BMS/BCU telemetry, health, alarm or status field.
negative_contactor_closed	bool	False	EMS-facing BMS/BCU telemetry, health, alarm or status field.
active_alarms	List[str]	factory/default list or dict	EMS-facing BMS/BCU telemetry, health, alarm or status field.
alarm_count	int	0	EMS-facing BMS/BCU telemetry, health, alarm or status field.
raw	Dict[str, Any]	factory/default list or dict	EMS-facing BMS/BCU telemetry, health, alarm or status field.

21. CSV Storage Schemas

The CSV schemas are defined in services/storage_logger.py. The Log API and database ingestion should treat these headers as the canonical historical log fields for the current CSV backend.

chiller_telemetry_header

#	Field	Meaning
1	timestamp	ISO timestamp for the sample or response.
2	sequence_no	Monotonic logger sequence number per logger instance.
3	gateway_id	Identifier of the gateway instance.
4	asset_id	Unique asset identifier, for example pcs_1, bms_1, chiller_1.
5	system_on_off	CSV field written by StorageLogger for this log type.
6	control_mode	CSV field written by StorageLogger for this log type.
7	set_temperature	CSV field written by StorageLogger for this log type.
8	outlet_water_temp	CSV field written by StorageLogger for this log type.
9	return_water_temp	CSV field written by StorageLogger for this log type.
10	outlet_water_pressure	CSV field written by StorageLogger for this log type.
11	return_water_pressure	CSV field written by StorageLogger for this log type.
12	ambient_temp	CSV field written by StorageLogger for this log type.
13	water_pump_status	CSV field written by StorageLogger for this log type.
14	compressor_1_status	CSV field written by StorageLogger for this log type.
15	compressor_2_status	CSV field written by StorageLogger for this log type.
16	electric_heater_status	CSV field written by StorageLogger for this log type.
17	condensate_fan_status	CSV field written by StorageLogger for this log type.
18	modbus_status	CSV field written by StorageLogger for this log type.

19	logger_status	CSV field written by StorageLogger for this log type.

pcs_telemetry_header

#	Field	Meaning
1	timestamp	ISO timestamp for the sample or response.
2	sequence_no	Monotonic logger sequence number per logger instance.
3	gateway_id	Identifier of the gateway instance.
4	asset_id	Unique asset identifier, for example pcs_1, bms_1, chiller_1.
5	vendor	Vendor/profile family.
6	comm_status	CSV field written by StorageLogger for this log type.
7	active_power_kw	CSV field written by StorageLogger for this log type.
8	reactive_power_kvar	CSV field written by StorageLogger for this log type.
9	apparent_power_kva	CSV field written by StorageLogger for this log type.
10	power_factor	CSV field written by StorageLogger for this log type.
11	frequency_hz	CSV field written by StorageLogger for this log type.
12	battery_voltage_v	CSV field written by StorageLogger for this log type.
13	battery_current_a	CSV field written by StorageLogger for this log type.
14	dc_power_kw	CSV field written by StorageLogger for this log type.
15	dc_total_current_a	CSV field written by StorageLogger for this log type.
16	bus_voltage_v	CSV field written by StorageLogger for this log type.
17	ab_voltage_v	CSV field written by StorageLogger for this log type.
18	bc_voltage_v	CSV field written by StorageLogger for this log type.
19	ca_voltage_v	CSV field written by StorageLogger for this log type.
20	phase_a_voltage_v	CSV field written by StorageLogger for this log type.
21	phase_b_voltage_v	CSV field written by StorageLogger for this log type.
22	phase_c_voltage_v	CSV field written by StorageLogger for this log type.
23	phase_a_current_a	CSV field written by StorageLogger for this log type.
24	phase_b_current_a	CSV field written by StorageLogger for this log type.
25	phase_c_current_a	CSV field written by StorageLogger for this log type.
26	operating_status	CSV field written by StorageLogger for this log type.
27	operating_status_raw	CSV field written by StorageLogger for this log type.
28	grid_offgrid_status	CSV field written by StorageLogger for this log type.
29	grid_offgrid_status_raw	CSV field written by StorageLogger for this log type.
30	fault_status	CSV field written by StorageLogger for

		this log type.
31	detailed_fault_status	CSV field written by StorageLogger for this log type.
32	fault_count	CSV field written by StorageLogger for this log type.
33	hardware_fault_word_1_raw	CSV field written by StorageLogger for this log type.
34	hardware_fault_word_2_raw	CSV field written by StorageLogger for this log type.
35	grid_fault_word_raw	CSV field written by StorageLogger for this log type.
36	bus_fault_word_raw	CSV field written by StorageLogger for this log type.
37	ac_capacitor_fault_word_raw	CSV field written by StorageLogger for this log type.
38	system_fault_word_raw	CSV field written by StorageLogger for this log type.
39	switch_fault_word_raw	CSV field written by StorageLogger for this log type.
40	other_fault_word_raw	CSV field written by StorageLogger for this log type.
41	active_faults	CSV field written by StorageLogger for this log type.
42	fault_words_read_error	CSV field written by StorageLogger for this log type.
43	igbt_temperature_c	CSV field written by StorageLogger for this log type.
44	ambient_temperature_c	CSV field written by StorageLogger for this log type.
45	inductance_temperature_c	CSV field written by StorageLogger for this log type.
46	error	CSV field written by StorageLogger for this log type.
47	logger_status	CSV field written by StorageLogger for this log type.

bms_telemetry_header

#	Field	Meaning
1	timestamp	ISO timestamp for the sample or response.
2	sequence_no	Monotonic logger sequence number per logger instance.
3	gateway_id	Identifier of the gateway instance.
4	asset_id	Unique asset identifier, for example pcs_1, bms_1, chiller_1.
5	communication_status	CSV field written by StorageLogger for this log type.
6	soc_percent	CSV field written by StorageLogger for this log type.
7	soh_percent	CSV field written by StorageLogger for this log type.
8	rack_inner_soc_percent	CSV field written by StorageLogger for this log type.
9	rack_voltage_v	CSV field written by StorageLogger for this log type.
10	rack_current_a	CSV field written by StorageLogger for this log type.
11	power_kw	CSV field written by StorageLogger for this log type.
12	max_allowed_charge_current_a	CSV field written by StorageLogger for this log type.
13	max_allowed_discharge_current_a	CSV field written by StorageLogger for

		this log type.
14	max_cell_voltage_mv	CSV field written by StorageLogger for this log type.
15	min_cell_voltage_mv	CSV field written by StorageLogger for this log type.
16	avg_cell_voltage_mv	CSV field written by StorageLogger for this log type.
17	cell_voltage_diff_mv	CSV field written by StorageLogger for this log type.
18	max_cell_temp_c	CSV field written by StorageLogger for this log type.
19	min_cell_temp_c	CSV field written by StorageLogger for this log type.
20	avg_temp_c	CSV field written by StorageLogger for this log type.
21	insulation_resistance_kohm	CSV field written by StorageLogger for this log type.
22	positive_insulation_resistance_kohm	CSV field written by StorageLogger for this log type.
23	negative_insulation_resistance_kohm	CSV field written by StorageLogger for this log type.
24	precharge_stage	CSV field written by StorageLogger for this log type.
25	bcu_state	CSV field written by StorageLogger for this log type.
26	current_state	CSV field written by StorageLogger for this log type.
27	positive_contactor_closed	CSV field written by StorageLogger for this log type.
28	precharge_contactor_closed	CSV field written by StorageLogger for this log type.
29	negative_contactor_closed	CSV field written by StorageLogger for this log type.
30	alarm_count	CSV field written by StorageLogger for this log type.
31	active_alarms	CSV field written by StorageLogger for this log type.
32	contactor_active_flags	CSV field written by StorageLogger for this log type.
33	last_error	Latest communication, storage or service error.
34	logger_status	CSV field written by StorageLogger for this log type.

chiller_event_header

#	Field	Meaning
1	timestamp	ISO timestamp for the sample or response.
2	gateway_id	Identifier of the gateway instance.
3	asset_id	Unique asset identifier, for example pcs_1, bms_1, chiller_1.
4	event_type	CSV field written by StorageLogger for this log type.
5	old_value	CSV field written by StorageLogger for this log type.
6	new_value	CSV field written by StorageLogger for this log type.
7	source	CSV field written by StorageLogger for this log type.
8	status	Status string such as ok, error, healthy, degraded, offline, disabled.
9	description	CSV field written by StorageLogger for

		this log type.

pcs_event_header

#	Field	Meaning
1	timestamp	ISO timestamp for the sample or response.
2	gateway_id	Identifier of the gateway instance.
3	asset_id	Unique asset identifier, for example pcs_1, bms_1, chiller_1.
4	vendor	Vendor/profile family.
5	event_type	CSV field written by StorageLogger for this log type.
6	command	Command string being executed.
7	old_value	CSV field written by StorageLogger for this log type.
8	new_value	CSV field written by StorageLogger for this log type.
9	readback_value	CSV field written by StorageLogger for this log type.
10	source	CSV field written by StorageLogger for this log type.
11	status	Status string such as ok, error, healthy, degraded, offline, disabled.
12	description	CSV field written by StorageLogger for this log type.
13	error	CSV field written by StorageLogger for this log type.

bms_event_header

#	Field	Meaning
1	timestamp	ISO timestamp for the sample or response.
2	gateway_id	Identifier of the gateway instance.
3	asset_id	Unique asset identifier, for example pcs_1, bms_1, chiller_1.
4	event_type	CSV field written by StorageLogger for this log type.
5	command	Command string being executed.
6	status	Status string such as ok, error, healthy, degraded, offline, disabled.
7	description	CSV field written by StorageLogger for this log type.
8	message	Human-readable response summary.
9	communication_status	CSV field written by StorageLogger for this log type.
10	soc_percent	CSV field written by StorageLogger for this log type.
11	rack_voltage_v	CSV field written by StorageLogger for this log type.
12	rack_current_a	CSV field written by StorageLogger for this log type.
13	precharge_stage	CSV field written by StorageLogger for this log type.
14	bcu_state	CSV field written by StorageLogger for this log type.
15	current_state	CSV field written by StorageLogger for this log type.
16	alarm_count	CSV field written by StorageLogger for this log type.
17	alarm	CSV field written by StorageLogger for

		this log type.
18	error	CSV field written by StorageLogger for this log type.

error_header

#	Field	Meaning
1	timestamp	ISO timestamp for the sample or response.
2	gateway_id	Identifier of the gateway instance.
3	asset_id	Unique asset identifier, for example pcs_1, bms_1, chiller_1.
4	error_type	CSV field written by StorageLogger for this log type.
5	error_source	CSV field written by StorageLogger for this log type.
6	description	CSV field written by StorageLogger for this log type.

22. Log Query and Filter Data Model

The log query layer is defined in core/storage/query. The LogFilter model decouples HTTP query parsing from CSV scanning so the same API contract can be preserved if storage later moves to SQLite, InfluxDB or cloud storage.

Field	Type	Default	Meaning
log_type	str		Log query/filter model field accepted by the query backend.
asset_id	Optional[str]	None	Log query/filter model field accepted by the query backend.
date	Optional[str]	None	Log query/filter model field accepted by the query backend.
start_time	Optional[str]	None	Log query/filter model field accepted by the query backend.
end_time	Optional[str]	None	Log query/filter model field accepted by the query backend.
fields	Optional[List[str]]	None	Log query/filter model field accepted by the query backend.
exact_filters	JsonDict		Log query/filter model field accepted by the query backend.
search	Optional[str]	None	Log query/filter model field accepted by the query backend.
limit	int	100	Log query/filter model field accepted by the query backend.
offset	int	0	Log query/filter model field accepted by the query backend.
order	str	'asc'	Log query/filter model field accepted by the query backend.

Filter group	Supported exact filter keys

Telemetry	bcu_state, comm_status, communication_status, current_state, fault_status, logger_status, modbus_status, operating_status, vendor
Events	command, event_type, source, status, vendor
Errors	error_source, error_type, source (source is accepted as alias for error_source)

Result field	Type	Default	Meaning
status	str		Returned in /api/logs/telemetry, /api/logs/events and /api/logs/errors responses.
rows	List[JsonDict]		Returned in /api/logs/telemetry, /api/logs/events and /api/logs/errors responses.
file	str	''	Returned in /api/logs/telemetry, /api/logs/events and /api/logs/errors responses.
file_name	str	''	Returned in /api/logs/telemetry, /api/logs/events and /api/logs/errors responses.
message	str	''	Returned in /api/logs/telemetry, /api/logs/events and /api/logs/errors responses.
total_rows	int	0	Returned in /api/logs/telemetry, /api/logs/events and /api/logs/errors responses.
filtered_rows	int	0	Returned in /api/logs/telemetry, /api/logs/events and /api/logs/errors responses.
rows_count	int	0	Returned in /api/logs/telemetry, /api/logs/events and /api/logs/errors responses.
limit	int	100	Returned in /api/logs/telemetry, /api/logs/events and /api/logs/errors responses.
offset	int	0	Returned in /api/logs/telemetry, /api/logs/events and /api/logs/errors responses.
filters	JsonDict		Returned in /api/logs/telemetry, /api/logs/events and /api/logs/errors responses.

23. BMS Register and Signal Dictionary

The following rows are extracted from drivers/bms_register_map.py. They document the BMS/BCU register-level source fields used to build EMS-facing BMS telemetry and control commands.

Group	Key	Address	Name	Category	Subcategory	Access	Format	Scale	Unit	Description
RACK_	function	0x10	Function	C6	Function	R	u16			Bitfield

SIGNAL	al_safety_warn		al Safety Warning	Battery Alarms & Faults	al Safety Faults					register for first-level functional safety faults.
RACK_SIGNAL	bcu_external_fault_alarm	0x11	BCU External Fault Alarm	C6 Battery Alarms & Faults	Critical Faults	R	u16			Bitfield register for BCU/BMU external critical faults.
RACK_SIGNAL	extern_critical_alarm	0x12	External Critical Alarm 1	C6 Battery Alarms & Faults	Critical Faults	R	u16			Bitfield register for first-level voltage/current/temperature/insulation faults.
RACK_SIGNAL	extern_alarm	0x13	External Alarm 1	C6 Battery Alarms & Faults	Second-Level Alarms	R	u16			Bitfield register for second-level voltage/current/temperature/insulation alarms.
RACK_SIGNAL	extern_warn	0x14	External Warning 1	C6 Battery Alarms & Faults	Third-Level Warnings	R	u16			Bitfield register for third-level warnings.
RACK_SIGNAL	extern_critical_alarm2	0x15	External Critical Alarm 2	C6 Battery Alarms & Faults	Critical Faults	R	u16			Bitfield register for additional first-level rack/module faults.
RACK_SIGNAL	extern_alarm2	0x16	External Alarm 2	C6 Battery Alarms & Faults	Second-Level Alarms	R	u16			Bitfield register for additional second-level rack/module alarms.
RACK_SIGNAL	extern_warn2	0x17	External Warning 2	C6 Battery Alarms & Faults	Third-Level Warnings	R	u16			Bitfield register for additional third-level

										rack/module warnings.
RACK_SIGNAL	pre_power_stage	0x18	Pre-Power / Precharge Stage	C1 Battery System Status	Precharge Status	R	u16			0=Idle, 1/2=Precharging, 3=Precharge success, 4=Precharge failed.
RACK_SIGNAL	bcu_state	0x19	BCU System State	C1 Battery System Status	System Operating Status	R	u16			0=Normal/Idle, 1=Charge forbidden, 2=Discharge forbidden, 3=Standby, 4=Stop/Shutdown.
RACK_SIGNAL	heart_beat_num	0x1a	Heartbeat Number	C1 Battery System Status	Communication Heartbeat	R	u16			Heartbeat frame/counter, expected to increment periodically.
RACK_SIGNAL	receive_cfg_state	0x1b	Receive Configuration State	C6 Battery Alarms & Faults	Vendor Extension	R	u16			0=BMU/slave registration failed, 1=BMU/slave registration success.
RACK_SIGNAL	pre_chg_fail_reason	0x1c	Precharge Failure Reason	C6 Battery Alarms & Faults	Vendor Extension	R	u16			Reserved/vendor-specific precharge failure reason.
RACK_SIGNAL	current_state	0x1d	Charge / Discharge State	C1 Battery System Status	Charge / Discharge State	R	u16			0=Idle, 1=Discharge, 2=Charge.
RACK_SIGNAL	full_or_empty_flag	0x1e	Full or Empty Flag	C1 Battery System Status	System Operating Status	R	u16			0=Full, 1=Empty, as per workbook. Field validatio

										n recommended.
RACK_SIGNAL	bcu_contactor_state	0x1f	BCU Contactor State	C1 Battery System Status	Contactor / Operating Status	R	u16			Bitfield: positive relay, precharge relay, negative relay, disconnector feedback.
RACK_MEASURE	max_cell_vol	0x210	Maximum Cell Voltage	C2 Battery Electrical Measurements	Cell Voltage Statistics	R	u16		mV	
RACK_MEASURE	max_cell_vol_num	0x211	Maximum Cell Voltage Number	C2 Battery Electrical Measurements	Cell Voltage Statistics	R	u16			
RACK_MEASURE	max_vol_bmu_num	0x212	Maximum Voltage BMU Number	C2 Battery Electrical Measurements	Cell Voltage Statistics	R	u16			
RACK_MEASURE	bmu_max_vol_num	0x213	BMU Maximum Voltage Cell Number	C2 Battery Electrical Measurements	Cell Voltage Statistics	R	u16			
RACK_MEASURE	min_cell_vol	0x214	Minimum Cell Voltage	C2 Battery Electrical Measurements	Cell Voltage Statistics	R	u16		mV	
RACK_MEASURE	min_cell_vol_num	0x215	Minimum Cell Voltage Number	C2 Battery Electrical Measurements	Cell Voltage Statistics	R	u16			
RACK_MEASURE	min_vol_bmu_num	0x216	Minimum Voltage BMU Number	C2 Battery Electrical Measurements	Cell Voltage Statistics	R	u16			
RACK_MEASURE	bmu_min_vol_num	0x217	BMU Minimum Voltage Cell Number	C2 Battery Electrical Measurements	Cell Voltage Statistics	R	u16			
RACK_MEASURE	max_cell_vol_diff	0x218	Maximum Cell Voltage Differenc	C2 Battery Electrical	Cell Voltage Statistics	R	u16		mV	

			e	Measurements						
RACK_MEASURE	ave_cell_vol	0x219	Average Cell Voltage	C2 Battery Electrical Measurements	Cell Voltage Statistics	R	u16		mV	
RACK_MEASURE	max_cell_temp	0x21a	Maximum Cell Temperature	C3 Battery Thermal Measurements	Thermal Measurements	R	s16	0.1	degC	
RACK_MEASURE	max_cell_temp_num	0x21b	Maximum Cell Temperature Number	C3 Battery Thermal Measurements	Thermal Measurements	R	u16			
RACK_MEASURE	max_temp_bmu_num	0x21c	Maximum Temperature BMU Number	C3 Battery Thermal Measurements	Thermal Measurements	R	u16			
RACK_MEASURE	bmu_max_temp_num	0x21d	BMU Maximum Temperature Number	C3 Battery Thermal Measurements	Thermal Measurements	R	u16			
RACK_MEASURE	min_cell_temp	0x21e	Minimum Cell Temperature	C3 Battery Thermal Measurements	Thermal Measurements	R	s16	0.1	degC	
RACK_MEASURE	min_cell_temp_num	0x21f	Minimum Cell Temperature Number	C3 Battery Thermal Measurements	Thermal Measurements	R	u16			
RACK_MEASURE	min_temp_bmu_num	0x220	Minimum Temperature BMU Number	C3 Battery Thermal Measurements	Thermal Measurements	R	u16			
RACK_MEASURE	bmu_min_temp_num	0x221	BMU Minimum Temperature Number	C3 Battery Thermal Measurements	Thermal Measurements	R	u16			
RACK_MEASURE	rack_ave_temp	0x222	Rack Average Temperature	C3 Battery Thermal Measurements	Thermal Measurements	R	s16	0.1	degC	
RACK_MEASURE	max_polarity_temp	0x223	Maximum Terminal Temperature	C3 Battery Thermal Measurements	Thermal Measurements	R	s16	0.1	degC	
RACK_MEASURE	max_polarity_num	0x224	Maximum Terminal Number	C2 Battery Electrical	Rack Measurements	R	u16			

				Measurements						
RACK_MEASURE	max_allowed_chg_cur	0x225	Maximum Allowed Charge Current	C2 Battery Electrical Measurements	Charge/Discharge Limits	R	u16	0.1	A	
RACK_MEASURE	max_allowed_dchg_cur	0x226	Maximum Allowed Discharge Current	C2 Battery Electrical Measurements	Charge/Discharge Limits	R	u16	0.1	A	
RACK_MEASURE	display_soc	0x227	Display SOC	C4 Battery Health & Capacity	SOC/SOH	R	u16	0.1	%	Workbook unit is per-mille; converted to percent by value * 0.1.
RACK_MEASURE	soh	0x228	SOH	C4 Battery Health & Capacity	SOC/SOH	R	u16	0.1	%	Workbook unit is per-mille; converted to percent by value * 0.1.
RACK_MEASURE	max_bat_temp_diff	0x229	Maximum Battery Temperature Difference	C3 Battery Thermal Measurements	Thermal Measurements	R	u16	0.1	degC	
RACK_MEASURE	max_bat_temp_rise	0x22a	Maximum Battery Temperature Rise	C3 Battery Thermal Measurements	Thermal Measurements	R	u16	0.1	degC	
RACK_MEASURE	max_bat_temp_rise_bmu_num	0x22b	Maximum Battery Temperature Rise BMU Number	C3 Battery Thermal Measurements	Thermal Measurements	R	u16			
RACK_MEASURE	max_bat_temp_rise_num_in_bmu	0x22c	Maximum Battery Temperature Rise Number In BMU	C3 Battery Thermal Measurements	Thermal Measurements	R	u16			
RACK_MEASURE	max_bat_temp_rise_num_in_bcu	0x22d	Maximum Battery Temperature Rise Number In BCU	C3 Battery Thermal Measurements	Thermal Measurements	R	u16			
RACK_	rack_inn	0x22e	Rack	C4	SOC/	R	u16	0.1	%	Workboo

MEASURE	er_soc		Internal SOC	Battery Health & Capacity	SOH					k unit is per-mille; converted to percent by value * 0.1.
RACK_MEASURE	pre_chg_vol	0x22f	Precharge Voltage	C2 Battery Electrical Measurements	Electrical Measurements	R	s16	0.1	V	
RACK_MEASURE	bmu_total_vol	0x230	BMU Total Voltage	C2 Battery Electrical Measurements	Electrical Measurements	R	s16	0.1	V	
RACK_MEASURE	total_vol	0x231	Total Rack Voltage	C2 Battery Electrical Measurements	Electrical Measurements	R	s16	0.1	V	
RACK_MEASURE	total_cur	0x232	Total Rack Current	C2 Battery Electrical Measurements	Electrical Measurements	R	s16	0.1	A	
RACK_MEASURE	ncon_total_vol	0x233	NCON Total Voltage	C2 Battery Electrical Measurements	Electrical Measurements	R	s16	0.1	V	
RACK_MEASURE	shunt_current	0x234	Shunt Current	C2 Battery Electrical Measurements	Electrical Measurements	R	s16	0.1	A	
RACK_MEASURE	can_hall_sample_cur_ori_value	0x235	CAN Hall Sample Current Original Value	C2 Battery Electrical Measurements	Electrical Measurements	R	s16	0.1	A	
RACK_MEASURE	cur_vol_hall_sam_cur_ori_value	0x236	Current/ Voltage Hall Sample Current Original Value	C2 Battery Electrical Measurements	Electrical Measurements	R	s16	0.1	A	
RACK_MEASURE	ir	0x237	Insulation Resistance	C5 Battery Safety & Protection	Insulation & Safety	R	u16		kOhm	
RACK_MEASURE	positive_ir	0x238	Positive Insulation Resistan	C5 Battery Safety & Protectio	Insulation & Safety	R	u16		kOhm	

			ce	n						
RACK_MEASURE	negative_ir	0x239	Negative Insulation Resistance	C5 Battery Safety & Protection	Insulation & Safety	R	u16		kOhm	
CONTROL	start_insulation_sampling	0x401	Start Insulation Sampling	C7 Battery Control Commands	Insulation Test Control	R/W	u16			Write 1 to start/reset insulation sampling; other value = no action.
CONTROL	start_precharge	0x402	Start Precharge Operation	C7 Battery Control Commands	Precharge Control	R/W	u16			Write 0x01 to start precharge; other value = stop/open precharge path.
CONTROL	bcu_reset	0x403	BCU Reset	C7 Battery Control Commands	Commissioning / Reset Control	R/W	u16			Write 0x01 to reset BCU; other value = no reset.
CONTROL	fan_switch	0x406	Fan Force Control	C7 Battery Control Commands	Fan Control	R/W	u16			Write 1=Fan ON, 2=Fan OFF. 0 may be used by EMS as auto/no-force if validated.

BMS control key	Command value mapping
start_insulation_sampling	{'start': 1, 'no_action': 0}
start_precharge	{'start': 1, 'stop': 0}
bcu_reset	{'reset': 1, 'no_action': 0}
fan_switch	{'auto': 0, 'on': 1, 'off': 2}

24. Database Integration Recommendation

The database team should not store raw frontend JSON blobs only. The recommended design separates asset metadata, current state, historical telemetry, event logs, error logs, health snapshots and command audit records.

Table	Primary fields	Purpose
assets	asset_id, asset_key, asset_type, vendor, protocol, profile, enabled, connection_json	Master asset catalog sourced from /api/assets

telemetry_current	asset_id, timestamp, status_json, selected_fields	Latest snapshot per asset for fast dashboard loading
telemetry_history	asset_id, timestamp, field_name, value_num, value_text, unit	Long-term timeseries for charts/analytics
events	asset_id, timestamp, event_type, command, source, status, old_value, new_value, description	Audit trail for commands and operational events
errors	asset_id, timestamp, error_type, error_source, description	Historical failures and diagnostics
health_snapshots	asset_id, timestamp, status, reason, recommended_action, connection_json, storage_json	Health monitoring trend and alerts
command_audit	request_id, asset_id, command, request_body_json, result_json, status, timestamp	Control/command traceability

25. Frontend and Backend Integration Guide

Screen/module	Recommended API	Notes
Dashboard home	GET /api/assets + GET /api/telemetry/operator	Discover assets dynamically and render operator-clean cards
Asset detail page	GET /api/assets/{asset_id} + GET /api/assets/{asset_id}/telemetry/operator	Show metadata, runtime mode, latest values and health
Live dashboard	GET /api/stream/telemetry?view=operator	Use SSE when browser live updates are required
Health page	GET /api/health/assets + GET /api/diagnostics	Show green/yellow/red health cards and recommended actions
Historical trend page	GET /api/assets/{asset_id}/telemetry/timeseries or Log API telemetry query	Use date/time/key filters for charts
Logs page	GET /api/logs/events, /api/logs/errors, /api/logs/telemetry	Expose field/date/time/search filters
Command panel	POST /api/assets/{asset_id}/commands	Use with confirmation prompts and display command result
Backend ingestion	GET /api/telemetry/latest or operator endpoint + /api/health	Store in database at 1-5 second interval depending on requirement

26. Example API Calls

Asset catalog

curl -s http://<IMX_WIFI_IP>:8000/api/assets | python3 -m json.tool

Operator telemetry

curl -s http://<IMX_WIFI_IP>:8000/api/telemetry/operator | python3 -m json.tool

PCS telemetry

curl -s http://<IMX_WIFI_IP>:8000/api/assets/pcs_1/telemetry/operator | python3 -m json.tool

Health

curl -s http://<IMX_WIFI_IP>:8000/api/health | python3 -m json.tool

Diagnostics

curl -s http://<IMX_WIFI_IP>:8000/api/diagnostics | python3 -m json.tool

Telemetry logs

curl -s "http://<IMX_ETH_IP>:7000/api/logs/telemetry?asset_id=pcs_1&date=YYYY-MM-DD&fields=timestamp,active_power_kw,dc_voltage_v&limit=20" | python3 -m json.tool

Command

```
curl -X POST http://<IMX_WIFI_IP>:8000/api/assets/pcs_1/commands -H "Content-Type: application/json" -d "{\"command\":\"PCS_READ\",
\"request_id\":\"web-001\"}"
```

27. Baseline Verification Checklist

- python -m compileall . passes without syntax errors.
- test_legacy_compatibility.py through test_log_filter_abstraction.py pass.
- Gateway starts with configs/actual_network_assets.json.
- Ports 6000, 7000 and 8000 are listening.
- /api/assets, /api/telemetry/operator, /api/health and /api/diagnostics work from browser/PowerShell.
- Log API filters work for date, time range, fields, search, events and errors.
- Flutter/Desktop receives UDP telemetry and TCP commands still work.
- PCS/BMS/chiller behavior remains hardware-verified for current deployment.

28. Future EMS Control Notes

The gateway baseline deliberately stops short of implementing EMS control state machines. That work should start only after the EMS control story, modes, asset interactions and safety policies are defined. The current baseline is ready because telemetry, command routing, health, logging and filters are stable and modular.